

The Changing Landscape of Care Work in Health in the Global South

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ABSTRACT

Care work is often moralized as essential to the functioning of society, but has long been undervalued economically and politically, contributing to a global care crisis and prompting increased control and extraction of care work to stabilize capitalist systems. The role of emerging technologies in these politics of care is understudied, particularly in the Global South. My research looks at why and how emerging technologies are being introduced into care work in health in India and Kenya, uncovering the implications this has for how care work is done, care workers' lives, and organizations deploying the technology. Drawing on cases of both harm and opportunity effected by technology, my work offers an empirical understanding of the role of technology in the politics of care work, and the implications this can have for the future of care work amidst a global care crisis.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; **Empirical studies in HCI**.

KEYWORDS

HCI4D, health, care work, future of work, India, Kenya

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1 CONTEXT AND MOTIVATION

As of 2018, the health and social work sector accounts for approximately 130 million jobs across the world, and more than two-thirds of these workers are women [2]. These jobs include community health workers, midwives, nurses, and doctors, and wages and working conditions for these jobs vary greatly according to geography, position in the healthcare hierarchy, privatization, and identity of the worker in terms of training and migration status. Countries in the geographic Global South in particular face shortages of health care workers, which has contributed to increased work burden, and

countries across the world underinvest in care sectors, further contributing to poor working conditions [2]. Thus, a global care crisis is arising, with the demand for paid care work increasing given the atomization of unpaid care work, aging populations, climate change, and the rise of epidemics and pandemics [2]. In response to this crisis and similar historic trends, we see the rise of ways to extract and control care work—for example through global care chains, where women from the Global South migrate for paid care work in order to financially support family at home [12].

There is a significant body of work in HCI on how technology is used by and could be better designed for healthcare workers. This work has largely been focused on nurses, doctors, and home care workers in the Global North [9], and community health workers in the Global South, looking at the design and introduction of technology such as electronic patient records and mobile apps (e.g., [7, 23, 33]). However, less work has sought to understand the role technology is playing in the politics of care more broadly, particularly in the Global South. Going beyond the role of technology in work, there are increasingly conversations on how the introduction of technology is reifying individualization of care [18], further justifying underinvestment in care [21], or taking advantage of the need to earn through care work [3, 27]. In the health domain in many Global South countries in particular, recent technological changes, prompted by organizational and global development priorities and increasing personal smartphone use among care workers and care recipients, have resulted in widespread yet underexplored impacts on care work and health infrastructures. Understanding these changes is essential to uncovering the impact of emerging technologies on mostly women care workers.

My dissertation work looks at how emerging technologies are shaping care work and care workers, situated in the politics of care and the push for technology in care work. My work looks at multiple contexts of technology use in India and Kenya, attending to why tech is introduced, how care workers at the bottom of the hierarchy do not always benefit from techno-optimist ideals, and how care workers are positioned in order to justify using technology in particular ways. My ongoing and future work then looks at a case of technology platforms intervening into existing care infrastructures and norms, opening up speculation around what role, if any, technology could play in an alternative, worker-centered politics of care.

2 KEY RELATED WORK

My work is at the intersection of studies of technology in care work, and conversations around the future of work.

A body of work, largely looking at the Global North, has looked at how technology can be better designed for care workers in health,

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including nurses and doctors [9], home health aides [30], non-clinical staff [1], and family caregivers [5, 11, 25]. Some of this work has engaged with the politics of feminized care work. Amidst the large body of work on electronic patient records, for example, some studies have looked at why they rarely meet nurses' needs and how the hierarchical relationships between workers contributes to invisible, additional, or unevenly distributed work [10, 28, 31]. Beyond HCI, conversations on the role of technology in the care crisis have focused on reproductive technologies and their role in surrogacy industries and fertility tourism [32]. Prior work has also found that introduction of technology into care work can also change standards of work and create more work—for example, the vacuum contributed to increased standards for cleanliness and required more labor from domestic workers overall to meet those standards [6].

In the Global South, a line of work has sought to understand how digital technologies might support community health workers as well as more formal health staff in training, data collection, and education (e.g., [7, 22–24, 33]), as well as trends in telehealth [4]. While these studies in Global South contexts engage with supporting care workers via technology, there is room to understand how sociotechnical systems shape and are shaped by the power dynamics that care work is often subjected to—particularly given the specificities that characterize different contexts, such as high patient-provider ratios and varying levels of informality of care work. Such analysis of feminized work in the Global South has been underexplored [29], and is just starting to grow. Recent work in India has brought attention to the power dynamics at play amidst increasing datafication and introduction of AI tools to community health workers' workflows [14]. Prior work has also looked at the platformization of wellness and beauty work in India and how platforms offer respectable modes of work but also take advantage of the need to earn [3, 27]. There is also a significant body of work that has looked at the transformation of labor other than care work, such as crowd work [13], gig work [19, 26], and job-seeking [8, 20], and which seeks to understand alternative organizing principles, such as collective power and safety nets [13, 29], towards more decent work. To these overlapping areas of research, I aim to contribute an understanding of the forces at play in the introduction of a range of technologies into healthcare work in Global South contexts, seeking to draw lessons from both harms and opportunities offered by technology.

3 RESEARCH QUESTIONS

My dissertation will answer the following interrelated questions, drawing on case studies across India and Kenya:

- How are emerging technologies shaping care work and care infrastructures in the geographic Global South?
- How do these sociotechnical systems interact with local politics of care?
- What do current transformations of work and healthcare infrastructures indicate about the future of gendered care work?

4 WORK TO DATE

In recent years, the technological landscape in the Global South has been characterized by increasing smartphone penetration, more affordable data, the transformation of social media and chat apps into platforms, and growing digital payment infrastructures. Thus, aspects of healthcare work such as payment, communication, and modes of delivery of care have been transformed. The studies described below look at case studies of these changes, contributing towards answering RQ1 and RQ2.

Study 1: *Digital payments in community health in rural Kenya.* In this study, I looked at how the introduction of digital payments, including bank payments and mobile money, was shaping the work and lives of community health workers in a county in southwest Kenya. This particular setting was also contextualized by the ongoing issue of missing payments from local governmental and non-governmental health organizations. I conducted semi-structured interviews with workers within the hierarchy of community health, and staff at the organizations that paid them. We found that many intersecting factors, such as the contracted and salaried status of health workers, digital and financial literacies, and location of residence, significantly shaped health workers' experience with digital payments. Community health volunteers (CHVs), who were at the bottom of the community health hierarchy, were not contracted or salaried, but did receive a monthly stipend at the behest of the local government. They incurred relatively heavy fees for the small amounts they were paid and faced challenges in withdrawing payments as cash to use in daily life. These challenges added to the burden of CHVs' unpaid care work and the contingencies of their paid work such as emergency expenses and local travel. Further, CHVs had little recourse for following up with missing payments and the inaccessibility of digital payment records made this even more difficult, particularly for workers with lower digital and financial literacies. Essential to the continuation of this dynamic was the positioning of CHVs by the government as volunteers who are simply appreciated with a stipend, as opposed to workers who are paid and can make demands. Notably, introduction of digital payments benefited the government and health organizations in this setting looking to disburse money more efficiently and easily. However, many workers' experiences contradicted the techno-optimist visions of digital payments as pathways to women's empowerment and highlighted the need for organizations to scaffold such changes in the financial landscape. This work was published at CSCW 2019 [15].

Study 2: *Chat in nurses' work in urban India.* In this study, I changed context to look at another shift in technology use that was more bottom-up—the widespread worker-driven adoption of chat apps in hospitals across the world, happening despite the many communication technologies specifically designed for hospitals. I aimed to understand how the use of WhatsApp was shaping nurses' work in a large hospital in urban South India. We conducted observations of nurses' work, semi-structured interviews with nursing and hospital management, and focus groups with nurses. We found that in a hierarchical and safety-critical workplace, the greater communication between staff nurses and their higher-ups facilitated by chat resulted in work for staff nurses that they had to take on off-shift. Once again, the way staff nurses were positioned by hospital

management contributed to the configurations of this sociotechnical system—part of the reason staff nurses were barred from using phones on shift was because of concerns that staff might become distracted or that they might share patient information, particularly of VIP patients. The hierarchical culture was also backdropped by the fact that many nurses relied on this work experience as a stepping stone to higher-paying nursing opportunities abroad. Despite democratic notions of worker-driven adoption of technology in prior work, we found that positionality within the care hierarchy shapes how much say workers have over their experience with technology. We also discuss how organizations seeking to formalize such communication tools in the workplace will need to account for the invisible work of chat. This study was published at CHI 2020 [17].

Study 3: *Chat-based peer support for youth living with HIV in urban Kenya.* Shifting to patient-provider communication, this study aimed to understand how increasing smartphone penetration and popularity of chat apps was shaping access to care. Chat-based health interventions reflect a larger shift towards remote health communication and pose the question of how this is affecting the work of care providers. In particular, I looked at the potential of chat as a platform for facilitated peer support for youth living with HIV. The study analyzed youth's and the facilitator's experience of chat-based peer support during a six-month pilot, using chat records and semi-structured interviews with youth and the facilitator. We found that chat-based peer support was successful in offering a space for youth to emotionally support one another and share experiential knowledge, in part due to the work of the facilitator. We found that the facilitator encouraged discussion, kept track of conversational threads, did the emotional labor of affirming and mediating between diverse perspectives, and ensured participants' access to the group despite issues like lost phones. These efforts fostered an environment in which youth could support one another and nurture their aspirations and the health practices required to achieve them. In addition to themes of privacy and aspirations in health, we discuss the design considerations for scaling such interventions, including how the work of the facilitator might be supported. This work received Best Paper at CHI 2021 [16].

5 ONGOING AND FUTURE WORK

My work so far has looked at multiple examples of how emerging technology is shaping care work and often exacerbating the challenging working conditions placed on care work. I now aim to understand how emerging technological changes are changing healthcare infrastructures in the Global South at a larger scale and what effect this is having on care work and care workers. Partnering with a health organization, my ongoing and future work addresses RQ3 by exploring the design, impact, and experience of a WhatsApp-based helpdesk for maternal and child health caregiving, deployed across multiple states in India. This reflects the increasing use of "chat for impact" in other countries such as South Africa and Bangladesh, indicating widespread trends in the future of care work that could be better understood.

The helpdesk I am studying is supported by an in-person component, where nurses at district hospitals give families trainings around maternal and child health caregiving. Families are then

given instructions on how to register for the helpdesk, through which they will receive automated informational messages and will have the opportunity to ask any questions about caregiving. The questions will be answered by trained nurses who support the helpdesk.

5.1 Research Approach

My ongoing and future work will use qualitative and design methods to understand workers' and families' experiences with the helpdesk. This work will be conducted in two phases, with Phase 1 currently ongoing and Phase 2 planned for 2022 (described below).

Phase 1 has been focused on gaining an in-depth, qualitative understanding of the design and implementation of the helpdesk, its use and impact, and nurses' work in supporting the helpdesk. Due to the pandemic, I have conducted online observations of nurses' work and analyzed chat records. I have also conducted semi-structured interviews with nurses, doctors, designers, and behavior change researchers at the health organization to understand the experience of conducting digital care work and the rationales, incentives, and constraints that have shaped the current design of the helpdesk. I am currently in the process of analyzing data from Phase 1. If possible, given the constraints of the pandemic, I also aim to expand Phase 1 by conducting interviews with families who use the helpdesk; the goal would be to understand the impact and experience of the helpdesk from their perspective, and how it is shaping unpaid care work.

5.2 Results

Initial analysis of Phase 1 data shows that the helpdesk presents a stark alternative to the disruption and burdening of care work seen in my prior work. This is partially due to how the helpdesk presents an alternative to physical nursing work. Compared to the wear of nursing work, the helpdesk offers nurses the opportunity to avoid night shifts, take care of family obligations, and work remotely. The health organization also relies on elevating care work in order to sustain its in-person component—nurses at district hospitals provide quality trainings on top of their regular workload partially because the health organization offers opportunities, such as attending conferences, and elevates nurses' status in the eyes of hospital administration.

The work required for supporting the helpdesk also positions care workers as tech workers in some ways, exhibiting a very different relationship with technology compared to findings in my prior work. In fact, a significant aspect of the work that the nurses enjoy is learning how to use new software, debugging logic behind the automated helpdesk replies, and creating reports to track analytics. This work also broadly aligns with what one nurse viewed as the future of healthcare work in India, in particular the increasing trends towards teleconsultations.

Based on chat records, the helpdesk seems to engage both mothers and fathers (when gender is indicated from the question phrasing). The helpdesk also offers a space to ask general informational questions and specific questions that come up about caregiving as care is being provided within the home. More outside of the expected scope of the helpdesk, however, were questions about how to navigate confusing or negative experiences with hospitals and

bureaucracy. There is the question of what forms of caregiving the system is funded and originally designed to support—that which takes place within the nuclear family. Though families are encouraged to share the information they learn through trainings with their neighbors, it may be important to ask how else the helpdesk can go beyond the “*low-hanging fruit*” (as one researcher at the organization noted) of individual and atomized behavior change within nuclear families.

5.3 Expected Next Steps

Based on data gathered from Phase 1, Phase 2 will explore plans for scaling that have been voiced by the organization, how this process will shape the work of nurses supporting the helpdesk, and what this indicates about trends in the future of care work in health organizations in the Global South. Phase 2 will continue to draw on semi-structured interviews to understand organizational definitions, goals, and requirements for scale, focused on the perspective of designers, researchers, and technical leads in the organization. Phase 2 will also include speculative design and co-design sessions with helpdesk nurses and other stakeholders (such as doctors) to understand what a worker-centered vision of scale might look like (and where scale is even appropriate).

6 CONTRIBUTIONS

My work will contribute to work in HCI and HCI for Development on the design of sociotechnical systems to better support care work, and on the future of work. Discourse on the future of work has only recently started looking at feminized sectors of work, despite long-growing concerns of a care crisis—deficits in care work despite increasing care needs. By focusing on feminized work and its situatedness in the political, economic, and social specificities of care in the geographic Global South, my work aims to contribute an understanding of the potential harms and opportunities that technology might have in this space. I also aim to consolidate principles of what a care worker-centered future of work looks like, not just through design of particular technologies, but through sociotechnical systems that prefigure alternative politics of care in which care work and care workers are valued.

7 BIO AND DISSERTATION STATUS

I am a PhD student in the Information and Communication Technologies and Development Lab at the University of Washington, advised by Richard Anderson. I am currently in my fifth year and my projected completion date is at the latest Spring 2023. I expect to conduct Phase 2 of my proposed work in the first half of 2022. The timing of the doctoral consortium would allow me to get feedback on the speculative and co-design components of the proposed work, and how to approach dissertation writing based on the range of studies I have conducted so far. After completion, I am aiming for research faculty positions in Information Schools or related departments.

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